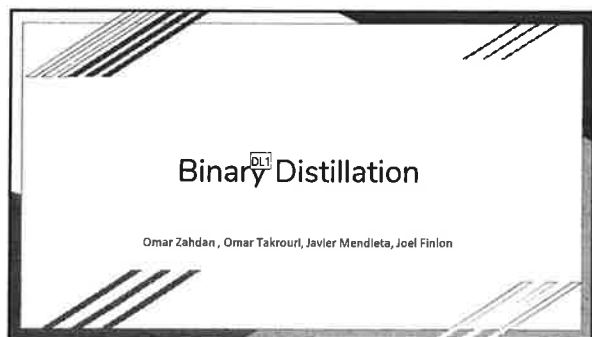
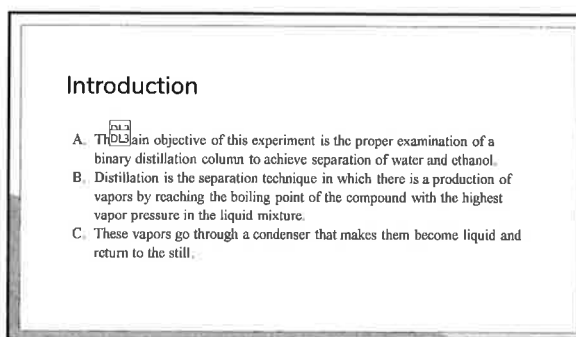
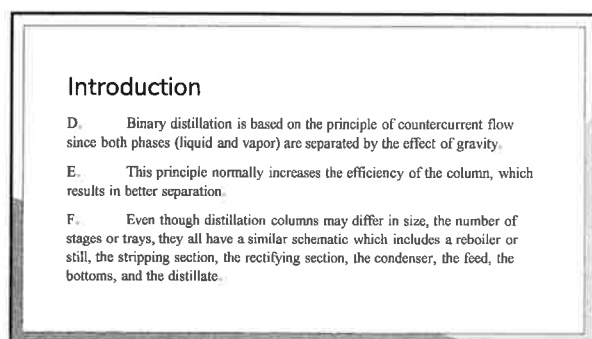




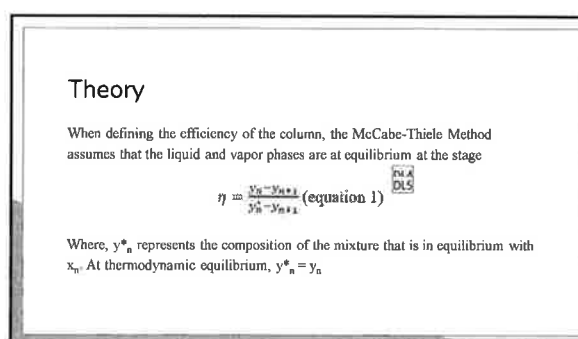
1



2



3



4

Slide 1

DL1 96/100
Donglin Li, 10/29/2019

Slide 2

DL2 Just the main objective is enough.
Donglin Li, 10/29/2019

DL3 -1
Donglin Li, 10/29/2019

Slide 4

DL4 Type the equation instead of pictures.
Donglin Li, 10/29/2019

DL5 -1
Donglin Li, 10/29/2019

Theory

The number of stages can be calculated by the following equation:

$$n = \frac{\ln \left(\frac{x_{d, \text{EtOH}}}{x_{b, \text{EtOH}}} \right) / \left(\frac{x_{d, \text{H}_2\text{O}}}{x_{b, \text{H}_2\text{O}}} \right)}{\ln \left(\frac{\alpha}{1} \right)} \quad (\text{equation 2})$$

In this equation, n corresponds to the number of theoretical plates, x_{EtOH} represents the mole fraction of ethanol (which is the more volatile compound in the mixture), $x_{\text{H}_2\text{O}}$ represents the mole fraction of water (which is the least volatile compound in the mixture), α corresponds to the average relative volatility. Furthermore, the d and b correspond to the labeling of distillate and bottoms, respectively.

5

Theory

The efficiency of the column can be calculated as follows:

$$\text{Efficiency} = \frac{\text{Number of theoretical plates}}{\text{Number of actual plates}} \times 100\% \quad (\text{equation 3})$$

6

Theory

In part B, the distillation column operates at a constant reflux ratio. This is carried out to determine how much the composition of the top and the bottom vary with time. This analysis requires that at constant total refluxes, the liquid and vapor flowing through the column are the same since there is no product and feed. Consequently, the following material balance can be applied to the McCabe-Thiele Method equations:

$$V = L \quad (\text{equation 4})$$

$$V y_n = L x_{n+1} \quad (\text{equation 5})$$

Which is further simplified to the following since V and L are equal to each other:

$$y_n = x_{n+1} \quad (\text{equation 6})$$

7

Apparatus

- 1) Reboiler
- 2) Bottom Product Receiver
- 3) Manometer
- 4) Column
- 5) Top Product Receiver
- 6) Condenser
- 7) Decanter
- 8) Top Product Collection Valve
- 9) Bottom Product Collection Valve



8

Apparatus

- 1) Reboiler - Heat a fraction of the bottom product into a vapor
- 2) Bottom Product Receiver - Vessel to collect product from the reboiler
- 3) Manometer - Measure pressure inside of column in mm H₂O
- 4) Column - The area where the liquid and vapor travel countercurrently to achieve separation
- 5) Top Product Receiver - Vessel to collect product from the condenser
- 6) Condenser - Unit where vapor is converted to liquid through a loss of heat
- 7) Decanter - Collects condensed vapor and moderates collection to top product receiver or total reflux
- 8) Top Product Collection Valve - Collection area to sample top product
- 9) Bottom Product Collection Valve - Collection area to sample bottom product

9

Material and Supplies

- 1) Ethanol and Distilled Water - used for calibration curve and for distillation (1st component)
- 2) DL9 Computer with UOP distillation column program - temperature profile
- 3) Heat Resistant Gloves - used when collecting bottom product
- 4) Test Tube - to collect bottoms
- 5) Parafilm - cover test tubes
- 6) Refractometer - device that measure refractive index

10

Material and Supplies

- 7) Latex Gloves - used throughout experiment
- 8) Beakers and graduated cylinders - used to collect and measure volume of collected sample.

11

Procedure

- Making 10% ethanol solution and loading in distillation column
- Making calibration mixtures of 10, 20, 30, etc. weight % ethanol and measuring refractive index
- Collecting overhead and bottom samples for ^{DL9} Experiment A
- Measuring pressure drop across column
- Collecting distillate samples for Experiment B
- Measuring refractive indices of final mixtures

12

Slide 10

DL8 Keep the styles the same when you itemize things.
Donglin Li, 10/29/2019

DL9 -1
Donglin Li, 10/29/2019

Slide 12

DL6 Be specific with Experiment A and Experiment B.
Donglin Li, 10/29/2019

DL7 -1
Donglin Li, 10/29/2019

Experimental Challenges

- Ensuring valves are fully closed to prevent leakage
- Calculating how much ethanol and water to mix for 10% ethanol mixture and calibration mixtures before the lab begins!
- Wait after switching boil-up rate to allow time for change
- Have someone on washing duty to speed up reuse of graduated cylinders

13

Safety

- Wear safety gloves at all times
- Wear safety goggles at all times
- Ensure all valves are fully closed! Ask TA to show you how they look when they are closed.
- Open valves slightly when collecting samples to avoid overflow and spillage

14

Thank you for your attention!
 ¡Gracias por vuestra atención!
 Merci pour votre attention !
 شكرا على الاهتمام!

15